Acute Diarrhea

By

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Definition and causes

Normal bowel frequency ranges from three times a day to three times a week in the normal population. Increased stooling, with stool consistency less solid than normal, ***Acute diarrhea*** is defined as a greater number of stools frequency or increased fluidity from the normal lasting for less than 14 days. If the illness persists for more than 14 days, it is called ***persistent***. If the duration of symptoms is longer than 1 month, it is considered ***chronic diarrhea***. Most cases of acute diarrhea are self limited, caused by infectious agents (e.g. viruses, bacteria, parasites), and do not require medication unless the patient is immunocompromised.

Prevalence

Worldwide, acute diarrhea constitutes a major cause of morbidity and mortality, especially among the very young, very old. Most cases are caused by enteric infections. It is also more prevalent in travelers to tropical regions; homosexual males, persons with underlying immunosuppression, and those living in nonhygienic environments, with exposure to contaminated water or foods.

Pathophysiology

Approximately 8 to 9 L of fluid enter the intestines daily—1 to 2 L represents food and liquid intake, and the rest is from endogenous sources such as salivary, gastric, pancreatic, biliary, and intestinal secretions. Most of the fluid, about 6 to 7 L, is absorbed in the small intestine, and only about 1 to 2 L is presented to the colon. Most of this is absorbed as it passes through the colon, leaving a stool output of about 100 to 300 g/day. Although many organisms simply impair the normal absorptive processes in the small intestine and colon, others, organisms, such as vibrio cholera, secrete a toxin that causes the colonic mucosa to secrete, rather than absorb, fluid and electrolytes. Voluminous diarrhea may result.

Causes of Acute Diarrhea

- Viral infections
- Bacterial infections
- Parasites
- Medicines
- Functional bowel disorder.

Viral Infections

- Rotavirus, which is the most common cause of diarrhea in children
- Adenovirus
- Norwalk virus
- Cytomegalovirus.
- Herpes simplex virus.

Bacterial Infections

Several types of bacteria can cause acute diarrhea. These include:

1- ***Shigella*** -- This bacterium is common in daycare settings and in rural villages in developing countries.

2-***Vibrio cholerae*** -- An infection with this bacterium (commonly referred to as cholera) is common in developing countries.

3-***Salmonella*** **species** -- A contaminant of eggs and poultry

4-***Yersinia enterocolitica*** -- A bacterium that causes an infection known as yersiniosis. This infection is caused by eating infected meat or dairy products

5- *Campylobacter jejuni* -- A contaminant of poultry. It is also found in household pets.

6- *Escherichia coli* (*E. coli*) -- Undercooked or raw hamburger has been implicated in many of the documented outbreaks of *E. coli*; however, outbreaks have also involved.

***7-Clostridium perfringens* --** A common contaminant in foods that have been warmed on steam tables

***8-Staphylococcus aureus* (*S. aureus*) -**A common contaminant of processed meats.

9- *Bacillus cereus* -- A common contaminant of rice and bean sprouts.

***10-Vibrio parahaemolyticus* --** An infection with this bacterium often results from eating raw seafood, especially oysters.

Parasitic infection

Parasites can enter the body through food or water and settle in the digestive system. Those that cause diarrhea include:

Giardia lamblia - A possible contaminant of drinking water, Giardia is also a common cause of diarrhea in homosexual men.

Entamoeba histolytica - This parasite is transmitted by fecal contamination of drinking water and foods, but also by direct contact with dirty hands or objects, as well as by sexual contact.

Cryptosporidium -- This parasite can be transmitted on any food touched by a contaminated food handler.

Medications

Some of these medicines include:

Antibiotics, Blood pressure medications,Laxatives,Antacids containing magnesium.

Other substances that may cause this condition include:

Alcohol ,Caffeine,Herbal teas.

Functional Conditions

The functional condition that most often results in acute diarrhea is anxiety and stress(Irritable bowel syndrome).

Symptoms and signs

Clinical features sometimes provide a clue as to the cause. Diarrhea caused by small intestine disease is typically high volume, watery, and often associated withmalabsorption. Dehydration is frequent. Diarrhea caused by **colonic involvement** is more often associated with frequent small-volume stools, the presence of blood, and a sensation of urgency.

Important factors in evaluating acute diarrhea :

-Travel history, sources of water (e.g., well water), recent food intake, history of profuse diarrheal episodes, dehydration, fever, hematochezia, nausea, vomiting, and abdominal pain. Important clinical features include abrupt versus gradual onset of symptoms, symptom duration, including bowel movement frequency, stool quantities, dysentery with fever, tenesmus, hematochezia or pus in the stool, signs of volume depletion, including thirst, tachycardia, orthostasis, decreased urine output, skin turgor, and lethargy or confusion, or both.

Patients ingesting **toxins** or those with toxigenic infection typically have nausea and vomiting as prominent symptoms, along with watery diarrhea, but rarely have a high fever. Vomiting that begins within 6 hours of ingesting a food should suggest food poisoning caused by preformed toxin from bacteria such as *S. aureus* or *B. cereus*. If diarrhea disease begins within 8 to 14 hours of food ingestion, *C. perfringens* should be suspected. When the incubation period is longer than 14 hours and vomiting is also a significant symptom, along with the diarrhea, viral agents should be considered. Parasites that do not invade the intestinal mucosa, such as *Giardia lamblia* and *Cryptosporidium*, usually cause only mild abdominal discomfort. Giardiasis may be associated with mild steatorrhea, gaseousness, and bloating.

Diagnosis

Acute diarrheas are usually infectious in origin and, for the most part, resolve with or without intervention before a diagnosis is made.

-The presence of blood is a useful clue, suggesting infection by invasive organisms, inflammation, ischemia, or neoplasm.

-Large-volume diarrhea suggests small bowel or proximal colonic disease, whereas small frequent stools associated with urgency suggest left colon or rectal disease. All current and recent medications should be reviewed, specifically new medications, antibiotics, antacids, and alcohol abuse.

A medical evaluation of acute diarrhea is not warranted in the previously healthy individual if symptoms are mild, moderate, spontaneously improve within 48 hours, and are not accompanied by fever, chills, severe abdominal pain, or blood in the stool

When to investigate :Evaluation is indicated if symptoms are severe or prolonged, the patient appears “toxic,” there is evidence of colitis (occult or gross blood in the stools, severe abdominal pain or tenderness, and fever), or empirical therapy has failed. Passage of many small-volume stools containing blood and mucus, temperature higher than 38.5° C (101.3° F), passage of more than six unformed stools in 24 hours, or a duration of illness longer than 48 hours, diarrhea with severe abdominal pain in a patient older than 50 years, diarrhea in older adults (older than 70 years) .

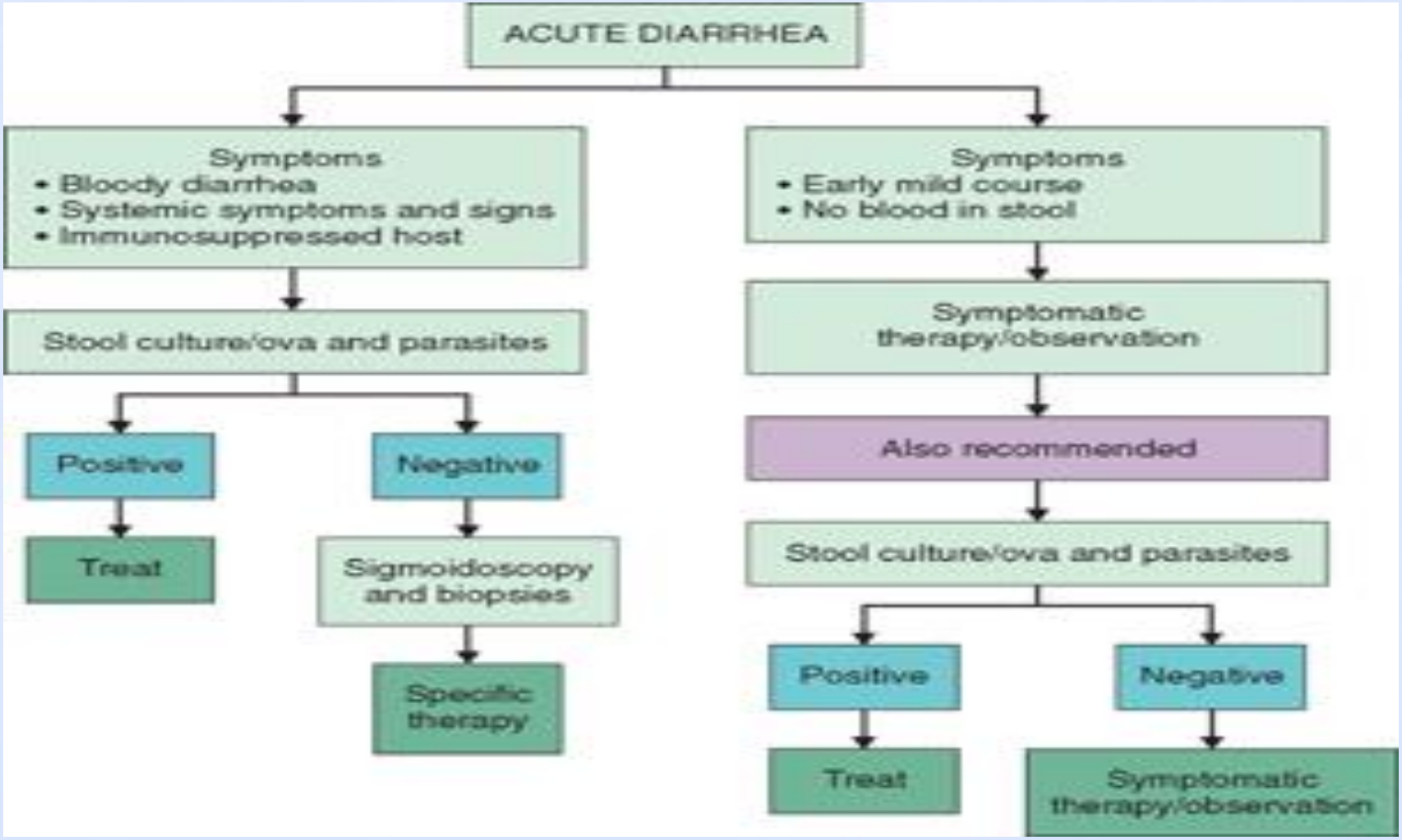
The physical examination Vital signs (including temperature and orthostatic evaluation of pulse and blood pressure) and signs of volume depletion (including dry mucous membranes, decreased skin turgor, and confusion) should be carefully evaluated. A careful abdominal examination to evaluate for tenderness and distention.

Stool analysis: Fecal leukocyte determination,Stool culture for enteric pathogens.Stool examination for ova and parasites

Stool evaluation for fecal leukocytes (or lactoferrin, a by-product of white blood cells) is a useful initial test, because it may support a diagnosis of inflammatory diarrhea. If the test is negative, stool culture may not be necessary, but culture is indicated if the test is positive.

Stool testing for ova and parasites Multiple fresh stool collections should be collected at different times because shedding of parasites may be intermittent.

A general algorithm for the evaluation and management of acute diarrhea is shown in Figure 1.



Flexible sigmoidoscopy with biopsy(sometimes it may be contraindicated).

Treatment: The principal components of the treatment of acute diarrhea are fluid and electrolyte replacement, dietary modifications, and drug therapy.

Rehydration. In most cases of acute diarrhea, fluid and electrolyte replacement are the most important forms of therapy.

Diet: Total food abstinence is unnecessary and not recommended. Foods providing calories are necessary to facilitate renewal of enterocytes. Patients should be encouraged to take well balanced diet. Dairy products should be avoided, because transient lactase deficiency can be caused by enteric, viral, and bacterial infections. Caffeinated beverages and alcohol, which can enhance intestinal motility and secretions, should be avoided.

Pharmacologic Measures:

Antidiarrheal Agents: These can be useful for the amelioration of symptoms. The most effective agents are the opioid derivatives—**Loperamide (Imodium**), **Diphenoxylate (Lomotil**)— and tincture of opium. These agents inhibit intestinal peristalsis, facilitating intestinal absorption, and have antisecretory properties. Loperamide may reduce the duration of diarrhea in those with traveler's diarrhea and bacillary dysentery. These agents should be avoided in patients with fever, bloody diarrhea, and possible inflammatory diarrhea because they may be associated with prolonged fever in patients with shigellosis, toxic megacolon in patients with C. difficile infection,

Bismuth subsalicylate,(Pepto-Bismol®, Kaopectate) somewhat less effective than loperamide, is effective in relieving symptoms of diarrhea, nausea, and abdominal pain in patients with traveler's diarrhea.

Antimicrobial Treatment:

- Because most patients have mild, self-limited disease caused by viruses or noninvasive bacteria, routine empirical treatment is not warranted. Empirical treatment is indicated for those patients with suspected invasive bacterial infection, traveler's diarrhea, or immunosuppression. Empirical antibiotic treatment is also appropriate specifically for early *Campylobacter* infections and *C. difficile*–associated diarrhea as well as for the febrile patient with fecal leukocytes and hemoccult-positive stools. Empirical treatment for *Giardia* can be prescribed for those with a 2-week or longer history of diarrhea.

Antibiotics may be recommended in certain situations, such as if you have the following signs or symptoms:

- Moderate to severe traveler's diarrhea
- More than eight loose stools per day, dehydration, symptoms that continue for more than one week, a weakened immune system, and in those who require hospitalization.

Specific therapy:Shigella or Salmonella speices: Quinolon therapy twice daily for 5 days.

Giardia lamblia infection: Metronidazol drug 250 mg tds for 7 days

Intestinal amoebiasia: Metronidazol 750mg t.i.d for 10 days.

EnterpathogenicE.coli: Quinolon therapy.

Capylobacter infection: Erythromycin 500mg for 5 days twice daily.

DIARRHEA PREVENTION

Food safety : Follow good hygienic measures.

Boil it,cookit,peal it or forget it

Summary

- Acute diarrhea is a common problem worldwide, with high morbidity and mortality in high-risk groups, such as the very young, older adults, and immunocompromised individuals.
- Most cases remit spontaneously and do not need diagnostic studies or specific therapy. Diagnostic studies are warranted for patients with fever or bloody diarrhea, or if the affected individual is immunocompromised.
- Infectious agents are responsible for most cases of acute diarrhea, and may act via numerous mechanisms.
- Prevention of dehydration is the most important therapeutic intervention for the management of individuals with acute diarrhea. Oral hydration is preferred in most cases.

- The use of nonspecific antidiarrheals, especially loperamide, is safe for most cases of acute diarrhea. Caution in use of these agents is urged when there is fever, chills, or bloody diarrhea, or when Clostridia difficile is suspected (diarrhea occurring after antibiotic use).
- Antibiotics such as ciprofloxacin may be used empirically for traveler's diarrhea. Antibiotics may also be used for early Campylobacter infection and for suspected or confirmed cases of C. difficile infection, as well as for the febrile patient who is positive for fecal leukocytes.

Case 1 A: 30 years female Pt. presented with diarrhea , tenesmus and rectal bleeding of 3 days duration . her motions were up to 20/ day and night. On examination she had temp 38.5oc and mild tenderness on the sigmoid colon.

The possible diagnosis is

Bacillary dysentery
Acute amoebic colitis
Bilharzial colitis
Cholera
Mention another possible diffirential diagnosis.

Case 2-An otherwise healthy 23-year-old man presents after the acute onset of watery diarrheathat has persisted for two days. He reports associated nausea and cramping butno emesis and is febrile, with a temperature of 38.7°C (101.7°F). How should he be evaluated and treated?

Suggested Questions:

1-Febrile diarrhea may be caused by the following organisms:
2-When u have to describe antibiotics to a case of acute diarrhea
3- Mention causes of bloody diarrhea.

Rifaximin (Rx):

Brand and Other Names:

Xifaxan

Classes:

Antibiotics, Other;

Antidiarrheals

Semisynthetics rifamycin based , non absorbable antibiotics that demonstrates no clinically relevant bacterial resistance

has very low rate of systemic

Has excellent safety profile

absorption 0.4%).

Dosage Forms & Strengths

tablet

200mg

550mg

Indications

Traveler's Diarrhea

Indicated for travelers' diarrhea caused by noninvasive strains of Escherichia coli (E coli)

200 mg PO q8hr for 3 days

Hepatic Encephalopathy

400mg PO q8hr for 7days.

Irritable Bowel Syndrome

Indicated for irritable bowel syndrome with diarrhea (IBS-D) in adult men and women

550 mg PO q8hr for 14 days; recurrence of symptoms can be retreated with a **14 day treatment course, up to 2 times**

Available in unit-dose package containing 42 tablets (ie, 2 week supply) for this indication.

Dosing Considerations

Limitations of use:

Not indicated or shown effective in patients with traveler's diarrhea complicated by fever or bloody stool or due to pathogens other than non-invasive strains of E coli

To reduce the development of drug-resistant bacteria and maintain effectiveness of antibiotics, rifamycin should be used only to treat or prevent infections that are proven or strongly suspected to be caused by bacteria .

CHRONIC DIARRHEA

Diarrhea is defined as recent change in bowel habits in the from of the presence of stool liquidity and an increase in daily stool weight, the upper normal limit of which is 200 grams in industrialized societies.

- Diarrhea is usually associated with increased stool frequency (more than three bowel movements per day). Diarrhea persistent for more than four weeks is considered chronic diarrhea.

PATHOPHYSIOLOGY OF DIARRHEA

- Osmotically active solutes in the intestinal lumen. (Osmotic diarrhea).
- Active ion secretion. (Secretory diarrhea)
- Deranged intestinal motility.
- Altered mucosal morphology or loss of absorptive surface.

(I) Osmotic Diarrhea

- Osmotic diarrhea is caused by the accumulation of non absorbed solutes in the gut lumen.
- Osmotic diarrhea stops when the patient fasts.
- The fecal fluid has a large solute gap, i.e., (Low electrolytes in stool).

Causes:

1. Ingestion of poorly absorbable solutes Magnesium sulfate, sodium sulfate, citrate-containing laxative! Some antacids—Mg(OH); Mannitol, sorbitol (chewing gum, diet candy)
2. Maldigestion ,Disaccharidase deficiencies, short bowel syndrome, postgastrectomy, postvagotomy state Chronic intestinal ischemia

(II) Secretory Diarrhea

- Stools are large in volume and watery and diarrhea persists with fasting. High electrolytes in stool

(small solute gap).

Causes :

- VIP production usually comes from tumors originating in pancreatic islet cells, vip-oma syndrome,
- Carcinoid syndrome
- Serotonin
- Medullary carcinoma of the thyroid
- Circulating calcitonin
- Zollinger-Ellison syndrome : - gastrin
- Bile acid diarrhea. ileal disease or resection.

(III) Deranged Intestinal Motility

- Abnormally reduced peristalsis may allow bacterial overgrowth in the small bowel.
- "Intestinal hurry".
- Premature emptying of the colon: irritable bowel syndrome, diabetic neuropathy, colonic tumour, fecal incontinence.
- The most frequent causes for sphincter incontinence are previous anal surgery for fissures, fistulas, or hemorrhoids; episiotomy or tear during childbirth; anal Crohn's disease

(IV) DIARRHEA DUE TO DISRUPTION OF STRUCTURAL INTEGRITY OF THE INTESTINE

- Sprue (tropical, nontropical).
- Whipple's disease
- Radiation enteritis
- Amyloidosis
- Collagen vascular diseases
- Inflammatory bowel disease
- Eosinophilic gastroenteritis Intestinal lymphoma
- Ileocecal tuberculosis Intestinal ischemia, mesenteric vasculitis
- Diverticulitis
- Acquired immunodeficiency syndrome, Giardia, Cryptosporidium.

(V) - MISCELLANEOUS CAUSES OF DIARRHEA

Drugs :

- Diuretics, cardiac glycosides, propranolol, colchicine, methotrexate, 6-mercaptopurine, 5-fluorouracil, guanethidine, ethanol
- Endocrine - :
- Addison's disease, hypoparathyroidism.
- Neurologic - :
- Tabestdorsalis, multiple sclerosis, myelitis, encephalitis, heat stroke, myotoniadystrophica, orthostatic hypotension.
- Toxins:
- Lead poisoning

MICROSCOPIC COLITIS.

Some patients with chronic diarrhea demonstrate inflammation of colonic mucosa despite a normal appearance of the colon on barium enema and colonoscopy.

- **History and Physical Examination**
  - When the stools are consistently large in volume, the underlying cause of diarrhea is likely to be located in the small bowel or in the proximal colon. By contrast, in small- volume diarrhea, in which the patient has frequent urges to defecate but passes only small amounts of feces or mucus, the disorder is usually in the left portion of the colon and rectum. Passage of blood mixed in with the diarrheal stool
  - Usually indicates inflammation of the mucosa, less often a neoplasm. Passage of non bloody mucus suggests irritable bowel syndrome, as does a history of small-volume diarrhea alternating with constipation. Frothy stools and excessive flatus suggest fermentation of unabsorbed carbohydrates.
  - Excessively foul stools suggest putrefaction of unabsorbed amino acids. Visible oil or fat indicates severe steatorrhea.
  - Fecal soiling (incontinence) suggests an anal sphincter defect.
  - Diarrhea in a patient with features of anorexia nervosa suggests laxative abuse.

There are, of course, many other pertinent facts obtainable from the history, including previous surgery, drug intake (Table 103-4), symptoms of systemic illness, travel, and related illnesses in family members. In chronic and recurrent diarrhea, an association of exacerbation of diarrhea with emotional stress should be sought, an association that may suggest irritable bowel syndrome.

❑ Clinical examination

The following signs may be associated

- Arthritis: Ulcerative colitis, Crohn's disease, Whipple's disease
- Liver disease: Ulcerative colitis, Crohn's disease, bowel malignancy with metastasis to liver
- Fever: Ulcerative colitis's disease, amebiasis, lymphoma, tuberculosis
- Marked weight loss:Malabsorption, inflammatory bowel disease, cancer, thyrotoxicosis
- ❑ Lymphadenopathy: Lymphoma, Whipple's disease, AIDS
- ❑ Neuropathy: Diabetic diarrhea, amyloidosis:
- ❑ Postural hypotension: Diabetic diarrhea, addison's disease,
- ❑ Flushing, large liver: Malignant carcinoid
- ❑ Pupura: Celiac disease

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- Rectal Biopsy. Amyloidosis, Whipple's disease, microscopic colitis.
- Quantitative Fecal Fat. Collected stools (usually for 72 hours)
- Twenty-four-how Stool Volume.

For example, stool volumes greater than 500 ml per day are rarely seen in patients with irritable bowel syndrome, and stool volumes of less than 1000 ml per day provide evidence against pancreatic cholera syndrome.

- Vasoactive Intestinal Polypeptide (VIP) and Other secretory hormones in blood.
- Therapeutic Trials.

These trials may include pancreatic enzymes, antibiotics, Metronidazole or anti-T.B drugs, cholestyramine, indomethacin (for prostaglandin synthetase inhibition), and various diets (lactose free, carbohydrate free, low fat, gluten free)

THERAPY

- The most satisfactory therapy is to cure the underlying disease.
- Supportive and symptomatic therapy

Opiates

Codeine, diphenoxylate with atropine (Lomotil), and loperamide

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